Part 1: Introduction to R

From a retail question to a reproducible answer · 307312 Applied Statistics

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- **Question:** Which product categories deserve a closer profitability review?
- **Data:** Tableau's fictitious Superstore; US orders placed in 2023–2025.
- **Outcome:** Import a CSV, inspect records, build a summary and explain a chart.
- Use this same retail dataset in Part 2 and the three practicals.

1.1 Set up your project

1. Install R, RStudio and Quarto.
2. Extract the complete student RStudio project.
3. Open `Applied-Statistics.Rproj`, then this notebook in `lessons/`.
4. Run cells in order. Choose **Render** to combine explanations, code and results.

- **R:** performs calculations.
- **RStudio:** provides an editor, Console, Environment and Plots pane.
- **Quarto:** produces the document you are reading.
- Keep useful code in the notebook so another analyst can reproduce your answer.

```
# Run once in the Console; do not reinstall every time you open a notebook.
install.packages(c("knitr", "rmarkdown", "dplyr", "ggplot2"))
```

1.2 Objects, vectors and missing values

Save and calculate

- `<-` saves a value; R distinguishes uppercase from lowercase.
- Quoted text is character data. `TRUE` and `FALSE` are logical values.
- The following values are **illustrative planning inputs**, not extra Superstore records.

```
unit_price <- 25
units <- 4
sales_value <- unit_price * units
sales_value
```

```
[1] 100
```

```
business_question <- "Which categories generate profit?"
sales_value > 80
```

```
[1] TRUE
```

Work with several values

```
# Illustrative order values, in dollars.
order_values <- c(25, 60, 100, 250)
mean(order_values)
```

```
[1] 108.75
```

```
median(order_values)
```

```
[1] 80
```

```
order_values[1]                   # R starts counting at 1
```

```
[1] 25
```

```
order_values[order_values >= 100]   # Keep entries meeting a condition
```

```
[1] 100 250
```

- `==`: equals; `!=`: differs; `&`: AND; `|`: OR; `%in%`: belongs to a set.
- `NA` means missing. It is different from zero.
- Check why a value is missing before excluding it.

```
values_with_missing <- c(25, 60, NA, 250)
is.na(values_with_missing)
```

```
[1] FALSE FALSE  TRUE FALSE
```

```
mean(values_with_missing)                # Missingness propagates
```

```
[1] NA
```

```
mean(values_with_missing, na.rm = TRUE)   # Mean of the observed values only
```

```
[1] 111.6667
```

```
sum(is.na(values_with_missing))
```

```
[1] 1
```

1.3 Import the retail data

- Download the order-line CSV.
- One row is **one product line within an order**.
- One order can have several lines. One customer can place several orders.
- Use the supplied CSV so your results match the course's pinned version.

```r
# This cell works from the project root or the lessons/ folder.
data_dir <- if (dir.exists("data")) "data" else file.path("..", "data")
source(file.path(data_dir, "..", "lessons", "_includes", "course-tools.R"))
lines <- read.csv(file.path(data_dir, "business", "superstore-lines.csv"),
                  stringsAsFactors = FALSE, colClasses = c(postal_code = "character"))
orders <- make_orders(lines)
c(order_lines = nrow(lines), orders = nrow(orders),
  customers = n_distinct(lines$customer_id))
```

```
order_lines      orders   customers
       6682        3322         782
```

- `read.csv()` creates a **data frame**: variables in columns, records in rows.
- `source()` loads the shared chart style and the explained order-summary function.
- `make_orders()` checks agreement of order attributes and sums sales, profit and units within each order.
- It calculates shipping time from order date to ship date; this is **not delivery time**.

```r
names(lines)
```

```
 [1] "row_id"        "order_id"      "order_date"    "ship_date"
 [5] "ship_mode"     "customer_id"   "customer_name" "segment"
 [9] "country"       "city"          "state"         "postal_code"
[13] "region"        "product_id"    "category"      "sub_category"
[17] "product_name"  "sales"         "quantity"      "discount"
[21] "profit"        "return_status"
```

```r
str(lines[c("order_id", "category", "sales", "quantity", "discount", "profit")])
```

```
'data.frame':   6682 obs. of  6 variables:
 $ order_id: chr  "US-2023-103800" "US-2023-112326" "US-2023-112326" "US-2023-112326" ...
 $ category: chr  "Office Supplies" "Office Supplies" "Office Supplies" "Office Supplies" ...
 $ sales   : num  16.45 3.54 11.78 272.74 19.54 ...
 $ quantity: int  2 2 3 3 3 9 2 3 2 4 ...
 $ discount: num  0.2 0.8 0.2 0.2 0.2 0 0 0 0 0 ...
 $ profit  : num  5.55 -5.49 4.27 -64.77 4.88 ...
```

```r
knitr::kable(head(lines[c("order_id", "category", "sales", "profit")]), digits = 2)
```

order_id	category	sales	profit
US-2023-103800	Office Supplies	16.45	5.55
US-2023-112326	Office Supplies	3.54	-5.49
US-2023-112326	Office Supplies	11.78	4.27
US-2023-112326	Office Supplies	272.74	-64.77
US-2023-141817	Office Supplies	19.54	4.88
US-2023-167199	Furniture	2573.82	746.41

```r
colSums(is.na(lines))
```

```
      row_id      order_id    order_date     ship_date     ship_mode
           0             0             0             0             0
 customer_id customer_name       segment       country          city
           0             0             0             0             0
       state   postal_code        region    product_id      category
           0             0             0             0             0
sub_category  product_name         sales      quantity      discount
           0             0             0             0             0
      profit return_status
           0             0
```

- Postal codes and IDs are labels, even when they contain digits.
- `discount = 0.20` means 20%, not 0.20%.
- Negative profit is a recorded loss, not automatically a data error.
- This extract has no missing cells; other datasets may do so.

1.4 Select, filter and create variables

- `select()` chooses columns.
- `filter()` chooses rows.
- `mutate()` adds or transforms variables.
- `%>%` passes the result of one step to the next.

```
loss_lines <- lines %>%
  filter(profit < 0) %>%
  select(order_id, category, sub_category, sales, discount, profit)
knitr::kable(head(loss_lines), digits = 2)
```

order_id	category	sub_category	sales	discount	profit
US-2023-112326	Office Supplies	Binders	3.54	0.8	-5.49
US-2023-112326	Office Supplies	Storage	272.74	0.2	-64.77
US-2023-105417	Furniture	Furnishings	76.73	0.6	-53.71
US-2023-105417	Office Supplies	Binders	10.43	0.8	-18.25
US-2023-118192	Office Supplies	Binders	3.44	0.7	-2.52
US-2023-149524	Furniture	Bookcases	61.96	0.5	-53.29

```
line_flags <- lines %>%
  mutate(loss_line = profit < 0,
         discount_pct = 100 * discount)
c(loss_lines = sum(line_flags$loss_line),
  loss_line_pct = 100 * mean(line_flags$loss_line))
```

```
  loss_lines loss_line_pct
  1251.00000      18.72194
```

- The denominator above is **6,682 order lines**.
- A loss-making line does not necessarily make the whole order unprofitable.

1.5 Group and summarize

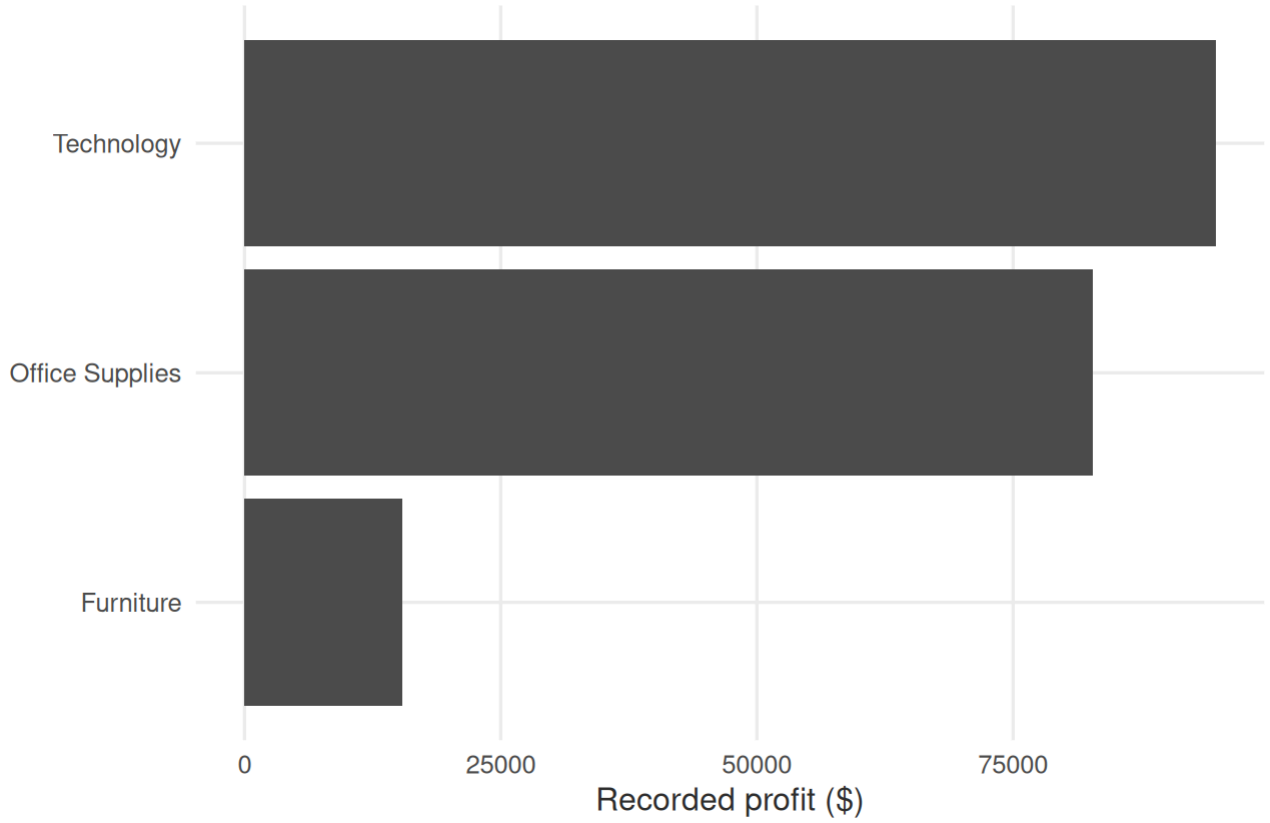
```
category_summary <- lines %>%
  group_by(category) %>%
  summarise(line_count = n(),
            total_sales = sum(sales),
            total_profit = sum(profit),
            .groups = "drop") %>%
  mutate(margin_pct = 100 * total_profit / total_sales) %>%
  arrange(desc(total_profit))
knitr::kable(category_summary, digits = 2)
```

category	line_count	total_sales	total_profit	margin_pct
Technology	1223	564423.2	94770.69	16.79
Office Supplies	4024	472949.9	82754.18	17.50
Furniture	1435	526612.5	15432.88	2.93

- `group_by()` defines which records belong together.
- `summarise()` returns one row per group.
- Overall margin is **total profit / total sales**.
- Averaging individual line margins gives a different, equally weighted statistic.

1.6 Make and interpret a chart

```
ggplot(category_summary, aes(x = reorder(category, total_profit), y = total_profit)) +
  geom_col(fill = petra_ink) + coord_flip() +
  labs(x = NULL, y = "Recorded profit ($)")
```



Recorded profit by category, US Superstore orders placed in 2023–2025.

- Furniture contributes much less profit than the other two categories in this extract.
- Compare its margin and subcategories before proposing an action.
- A chart identifies a pattern; it does not establish why the pattern occurred.

1.7 Your first business briefing

1. Summarize total sales and total profit by `region`.
2. Calculate each region's overall profit margin.
3. Draw a bar chart answering one clear question.
4. Write three bullets: a finding, supporting numbers and a useful next investigation.

```
# Start with lines, group by region, then summarize and calculate a margin.
# Explain what each row of your result represents.
```

1.8 Before moving on

- Can you explain the difference between a line, an order and a customer?
- Can you identify the denominator of a percentage?
- Can another student rerun your code from the complete project folder?
- Continue to Part 2.

Source and data notes

- Tableau Superstore sample data: a fictitious company, not audited company accounts.
- Tableau's explanation of row detail.
- The course uses a fixed US, 2023–2025 extract from the workbook downloaded on 6 November 2026.
- Full variables, units, selection rules and limitations: course data guide.